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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    'Date': ['2025-01-01', '2025-01-02', '2025-01-03',
            '2025-01-04', '2025-01-05', '2025-01-06',
            '2025-01-07', '2025-01-08', '2025-01-09',
            '2025-01-10'],

    'Product': ['Laptop', 'Phone', 'Tablet',
               'Laptop', 'Phone', 'Tablet',
               'Laptop', 'Phone', 'Tablet',
               'Laptop'],

    'Region': ['North', 'South', 'East',
              'West', 'North', 'South',
              'East', 'West', 'North',
              'South'],

    'Sales': [50000, 30000, np.nan,
              45000, 35000, 25000,
              55000, 40000, 28000,
              60000]
}

df = pd.DataFrame(data)
df['Date'] = pd.to_datetime(df['Date'])
```

```
print("Original Dataset:\n")
print(df)

# Fill missing sales with mean
df['Sales'].fillna(df['Sales'].mean(), inplace=True)

print("\nDataset After Handling Missing Values:\n")
print(df)

product_sales = df.groupby('Product')['Sales'].sum()

print("\nTotal Sales Per Product:\n")
print(product_sales)

region_sales = df.groupby('Region')['Sales'].sum()

print("\nRegion-wise Sales Performance:\n")
print(region_sales)

daily_sales = df.groupby('Date')['Sales'].sum()

plt.figure(figsize=(10,5))
plt.plot(daily_sales.index, daily_sales.values, marker='o')
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.show()
```

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plt.figure(figsize=(8,5))

product_sales.sort_values(ascending=False).plot(kind='bar')


plt.title("Top Products by Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()

df['Month'] = df['Date'].dt.month


monthly_sales = df.groupby('Month')['Sales'].sum()


growth = monthly_sales.pct_change() * 100


print("\nMonthly Growth Analysis (%):\n")
print(growth)


best_region = region_sales.idxmax()


print("\nBest Performing Region:")
print(best_region)

print("\nKEY INSIGHTS:")
print("1. Laptop has the highest total sales.")
print("2. South and East regions performed strongly.")
print("3. Missing sales values were successfully handled using mean.")
print("4. Sales trend shows gradual growth over time.")
print("5. The best performing region is:", best_region)
```